

July 1, 2026

Dr. Mohammad Soorati
School of Electronics & Computer Science
University of Southampton

Dear Dr. Mohammad Soorati,

I am writing to express my strong interest in the Savvas Chamberlain Scholarship PhD position focusing on using machine learning to model complex biological interactions. Having recently graduated as an AI & Data Science Engineer specializing in Physics-Informed Machine Learning and continuous-time series modeling, I am highly motivated to transition my expertise into dynamic biological computer vision.

During my Master's thesis ("Cardiovascular Medical Digital Twin"), my core research focused on modeling continuous, temporal biological dynamics from highly irregular data. I engineered Medical_MambaTS, a state-space model capable of real-time clinical sequence modeling, and PI-DeepONet, a Physics-Informed Operator Network that maps the exact differential equations of hemodynamics into the PyTorch Autograd graph. I understand that traditional experimental snapshots fail to capture the behavioral dynamics of immune cells attacking cancer cells. I believe the continuous-time sequence modeling and spatial-temporal architectures I developed for digital twins are directly transferable to analyzing time-lapse microscopy videos.

Furthermore, my State Engineer thesis ("Gas Turbine Digital Twin") required me to build Spatial-Temporal Transformers that processed high-frequency multidimensional telemetry to identify fault patterns that humans could not observe. Applying these spatial-temporal sequence models to track the dynamic interactions between immune and malignant cells represents the exact intersection of Explainable AI and complex systems biology that I wish to pursue for my PhD.

I possess extensive programming skills in Python, PyTorch, and Computer Vision frameworks. Please find attached my CV, academic references, research proposal, and the one-page summaries of both my Engineering and Master's theses. I would welcome the opportunity to discuss my background and ideas with you in an interview.

Sincerely,
Ismail Aissaoui
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